Biochemistry Cume
Feb 10, 2006

Nucleic Acids (announced).

1. (20 points) Complete each sentence.

    a. The chemical site for elongation of a DNA polymer is: _____ .
    b. The chemical site for elongation of an RNA polymer is: _____ .
    c. One of the monomers used for elongation of a DNA polymer is: _____ .
    d. One of the monomers used for elongation of an RNA polymer is: _____ .

2. (10 points) Draw the chemical structure of your answer to 1(d).

3. (15 points) Use a combination of chemical and biological arguments to explain why T and not U is found in natural sources of DNA.

4. (15 points) An aqueous solution of DNA is stored overnight at pH 12.
Give a detailed chemical mechanism for any significant reaction that occurs.

5. (15 points) An aqueous solution of RNA is stored overnight at pH 12.
Give a detailed chemical mechanism for any significant reaction that occurs.

6. (10 points) Define each term.

    a. Restriction enzyme.
    b. Ligase.

7. (15 points) Outline a convenient general method for the *in vitro* or *in vivo* amplification of DNA from a fossil source. Describe essential steps in your protocol.